

# Expenses Reporting Tool for an IT Consultancy

Project Management (GEP)  
Assignment 4: Final Document

Bachelor Degree in Informatics Engineering  
Specialization in Software Engineering

Bachelor's Thesis



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April 10, 2026

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## 6 Budget and Sustainability

Before closing the initial project documentation, a detailed analysis of its economic viability and sustainability across economic, environmental, and social dimensions is provided. The budget is based on tasks and resources defined in Section ??, while the sustainability assessment evaluates the long-term impact of the proposed solution.

### 6.1 Budget

This section details all project expenses, justifying each cost and its corresponding calculation. The budget is structured into four main categories: Personnel Cost (PC), Generic Cost (GC), Contingency Cost (CC), and Incidental Cost (IC).

#### 6.1.1 Personnel Cost

The personnel cost is calculated from the effort estimates presented in Table ?. To obtain a realistic estimate, the average hourly wage for each personnel role was calculated in advance using market salary data[1], assuming 260 working days per year and an average of 7.5 working hours per day.

Table 6.1.1 specifies the hourly wage calculation for each role, including the Social Security (SS) cost. Each human resource identified in Section ?? is assigned to a specific role in the table, resulting in 5 different roles:

- **Project Manager (PM)**: Responsible for planning and leading the entire project.
- **Junior Full-Stack Developer (D)**: Responsible for the full application implementation
- **QA Tester (T)**: Responsible for verifying the application functionalities.
- **Software Architect (SA)**: Responsible for designing the application's technical architecture while considering the project requirements and scope.
- **Technical Writer (TW)**: Responsible for drafting clear, accurate project documentation.

| Role                            | Annual Salary (€) | Total including SS (€) | Cost / Hour (€) | Assigned HR |
|---------------------------------|-------------------|------------------------|-----------------|-------------|
| Project Manager (PM)            | 43,300            | 56,290                 | 28.87           | PO, TD, GT  |
| Full-stack Junior Developer (D) | 29,800            | 38,740                 | 19.87           | DT          |
| QA Tester (T)                   | 26,888            | 34,954.4               | 17.93           | DT          |
| Software Architect (SA)         | 41,800            | 54,340                 | 27.87           | DT          |
| Technical Writer (TW)           | 31,300            | 40,690                 | 20.87           | DT          |

Table 6.1.1: Annual and hourly salaries for different project roles. [Own creation]

Given each personnel member's hourly wage and their corresponding effort hours in each task, the total personnel cost is calculated as shown in Table 6.1.2. Each task cost results from multiplying both values.

| ID  | Name                            | Total hours | Hours per role |      |   |    |    |          | Cost (€) |
|-----|---------------------------------|-------------|----------------|------|---|----|----|----------|----------|
|     |                                 |             | PM             | D    | T | SA | TW |          |          |
| PS  | Project Starting                | 50          | 80             | 0    | 0 | 0  | 0  | 1,433.33 |          |
| PS1 | Context and Scope               | 20          | 20             | 0    | 0 | 0  | 0  | 577.33   |          |
| PS2 | Temporal Planning               | 10          | 10             | 0    | 0 | 0  | 0  | 288.67   |          |
| PS3 | Budget and Sustainability       | 10          | 10             | 0    | 0 | 0  | 0  | 288.67   |          |
| PS4 | Initial Documentation           | 10          | 10             | 0    | 0 | 0  | 0  | 288.67   |          |
| PM  | Project Management              | 46.5        | 38.5           | 46.5 | 0 | 0  | 0  | 2,035.17 |          |
| PM1 | Product Owner Meetings          | 30.5        | 30.5           | 30.5 | 0 | 0  | 0  | 1486.37  |          |
| PM2 | Thesis Director Review Meetings | 8           | 8              | 8    | 0 | 0  | 0  | 389.87   |          |

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| ID           | Name                                                           | Total hours | PM          | D            | T         | SA        | TW          | Cost (€)         |
|--------------|----------------------------------------------------------------|-------------|-------------|--------------|-----------|-----------|-------------|------------------|
| PM3          | Product Backlog Update                                         | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| <b>DT</b>    | <b>Design</b>                                                  | <b>16</b>   | <b>0</b>    | <b>0</b>     | <b>0</b>  | <b>16</b> | <b>0</b>    | <b>445.87</b>    |
| DT1          | Use Case Diagram                                               | 2           | 0           | 0            | 0         | 2         | 0           | 55.73            |
| DT2          | Conceptual Model Diagram                                       | 4           | 0           | 0            | 0         | 4         | 0           | 111.47           |
| DT3          | Technology Stack Decision                                      | 2           | 0           | 0            | 0         | 2         | 0           | 55.73            |
| DT4          | Logical and Physical Architecture Design                       | 4           | 0           | 0            | 0         | 4         | 0           | 111.47           |
| DT5          | UI mockups Design                                              | 2           | 0           | 0            | 0         | 2         | 0           | 55.73            |
| DT6          | Security Design                                                | 2           | 0           | 0            | 0         | 2         | 0           | 55.73            |
| <b>FPD</b>   | <b>Final Product Delivery</b>                                  | <b>56.5</b> | <b>0</b>    | <b>5</b>     | <b>0</b>  | <b>0</b>  | <b>51.5</b> | <b>1173.97</b>   |
| FPD1         | User Manual Documentation                                      | 2           | 0           | 1            | 0         | 0         | 1           | 40.73            |
| FPD2         | Final Thesis Documentation                                     | 30.5        | 0           | 0            | 0         | 0         | 30.5        | 636.43           |
| FPD3         | Final Demonstration Video                                      | 4           | 0           | 4            | 0         | 0         | 0           | 79.47            |
| FPD4         | Bachelor's Thesis Defense                                      | 20          | 0           | 0            | 0         | 0         | 20          | 417.33           |
| <b>Dev</b>   | <b>Development</b>                                             | <b>86</b>   | <b>0</b>    | <b>76</b>    | <b>10</b> | <b>0</b>  | <b>0</b>    | <b>1689.12</b>   |
| Dev0         | Self-learning and study of unfamiliar technologies             | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| Dev1         | Environment Configuration                                      | 2           | 0           | 2            | 0         | 0         | 0           | 39.73            |
| Dev2         | Product Deployment                                             | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| Dev3         | CI/CD Pipeline Implementation                                  | 20          | 0           | 10           | 10        | 0         | 0           | 377.92           |
| Dev4         | Logging and Monitoring                                         | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| Dev5         | Git Repository Documentation                                   | 40          | 0           | 40           | 0         | 0         | 0           | 794.67           |
|              | <b>Frontend and Backend Development</b>                        | <b>128</b>  | <b>0</b>    | <b>128</b>   | <b>0</b>  | <b>0</b>  | <b>0</b>    | <b>2,542.93</b>  |
| <b>ER</b>    | <b>Expense and Report Development</b>                          | <b>64</b>   | <b>0</b>    | <b>64</b>    | <b>0</b>  | <b>0</b>  | <b>0</b>    | <b>1,271.47</b>  |
| ER1          | Expense Uploads                                                | 16          | 0           | 16           | 0         | 0         | 0           | 317.87           |
| ER2          | Uploaded Expenses Real-time Dashboard                          | 4           | 0           | 4            | 0         | 0         | 0           | 79.47            |
| ER3          | AI-Extracted Expense Data Review and Correction                | 20          | 0           | 20           | 0         | 0         | 0           | 397.33           |
| ER4          | Uploaded Expenses Deletion                                     | 4           | 0           | 4            | 0         | 0         | 0           | 79.47            |
| ER5          | Expense Reports View, Creation, Edit, Submission, and Deletion | 20          | 0           | 20           | 0         | 0         | 0           | 397.33           |
| <b>UA</b>    | <b>User Access Development</b>                                 | <b>64</b>   | <b>0</b>    | <b>64</b>    | <b>0</b>  | <b>0</b>  | <b>0</b>    | <b>1,271.47</b>  |
| UA1          | Company Email User Authentication                              | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| UA2          | Role-based User Access                                         | 16          | 0           | 16           | 0         | 0         | 0           | 317.87           |
| UA3          | Submitted Report Approval and Rejection                        | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| UA4          | Employee Spending Limit Configuration                          | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| UA5          | Submitted and Reviewed Expenses Record Visualization           | 4           | 0           | 4            | 0         | 0         | 0           | 79.47            |
| UA6          | Email Notification Service Integration                         | 20          | 0           | 20           | 0         | 0         | 0           | 397.33           |
| <b>BD</b>    | <b>Backend Development</b>                                     | <b>120</b>  | <b>0</b>    | <b>120</b>   | <b>0</b>  | <b>0</b>  | <b>0</b>    | <b>2,384.00</b>  |
| BD1          | RESTful API Backend Logic Implementation                       | 20          | 0           | 20           | 0         | 0         | 0           | 397.33           |
| BD2          | Azure SQL Database Integration and Manipulation                | 20          | 0           | 20           | 0         | 0         | 0           | 397.33           |
| BD3          | Azure Blob Storage Integration                                 | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| BD4          | Microsoft Document Intelligence AI Model Integration           | 20          | 0           | 20           | 0         | 0         | 0           | 397.33           |
| BD5          | Automated AI-driven Data Extraction Pipeline                   | 20          | 0           | 20           | 0         | 0         | 0           | 397.33           |
| BD6          | Azure SignalR Real-time Notification Service Integration       | 16          | 0           | 16           | 0         | 0         | 0           | 317.87           |
| BD7          | Microsoft Entra ID User Authentication Integration             | 16          | 0           | 16           | 0         | 0         | 0           | 317.87           |
| <b>FD</b>    | <b>Frontend Development</b>                                    | <b>10</b>   | <b>0</b>    | <b>10</b>    | <b>0</b>  | <b>0</b>  | <b>0</b>    | <b>198.67</b>    |
| FD1          | Expenses Search and Filter                                     | 8           | 0           | 8            | 0         | 0         | 0           | 158.93           |
| FD2          | Reports Search                                                 | 2           | 0           | 2            | 0         | 0         | 0           | 39.73            |
| <b>Total</b> |                                                                | <b>513</b>  | <b>88.5</b> | <b>385.5</b> | <b>10</b> | <b>16</b> | <b>51.5</b> | <b>11,913.05</b> |

Table 6.1.2: Personnel cost breakdown by task, role, and total personnel cost. [Own creation]

### 6.1.2 Generic Cost

This category covers the project operational costs and is based on the resources identified in Section ???. The project duration, calculated in advance in Section ?? (513 hours over 122 days), is used in each cost calculation.

#### Amortization

This category cost corresponds to the proportion of the hardware resources usage time compared to the total usable lifespan. Hence, the amortization cost is formulated as:

$$\text{Amortization (€)} = \frac{\text{Resource Price}}{\text{Total Depreciation Hours}} \times \text{Project Usage Hours}$$

The usage hours equal the project execution time, since hardware resources are used throughout the project. Assuming a four-year depreciation period and the same annual working hours from PC (260 working days \* 7.5 working hours/day), which is a total of 1950 hours, the following equation is applied for each hardware resource in Table 6.1.3:

$$\text{Amortization (€)} = \text{Resource price} \times \frac{1}{4 \text{ years}} \times \frac{1 \text{ year}}{1950 \text{ working hours}} \times \text{hours used}$$

| Hardware Resources | Price (€) | Quantity | Total Price (€) | Usage Time (h) | Amortization (€) |
|--------------------|-----------|----------|-----------------|----------------|------------------|
| PC Dell 16 Pro [2] | 1,193.53  | 1        | 1,193.53        | 513            | 78.50            |
| Wireless Mouse [3] | 11.99     | 1        | 16.99           | 513            | 1.12             |
| Dell Monitor [4]   | 171.82    | 2        | 343.64          | 513            | 22.60            |
| Keyboard [5]       | 19.99     | 1        | 19.99           | 513            | 1.31             |
| <b>Total</b>       |           |          |                 |                | <b>102.22</b>    |

Table 6.1.3: Amortization cost of the project's hardware resources. [Own creation]

### Electric cost

Only hardware resources require electricity and are the only resources considered in this cost. Electricity costs are calculated by multiplying the current electricity cost of **0.1748 €/kWh** [6] by the energy consumed and the resource usage time. Table 6.1.4 details the calculations and the resulting total cost:

| Hardware Resources | Power (W) | Usage Time (h) | Energy Consumed (kWh) | Electricity Cost (€) |
|--------------------|-----------|----------------|-----------------------|----------------------|
| PC Dell 16 Pro [2] | 65        | 513            | 33.345                | 5.83                 |
| Wireless Mouse [3] | 0.05      | 513            | 0.02565               | 0.00                 |
| Dell Monitor [4]   | 24        | 513            | 12.312                | 2.15                 |
| Keyboard [5]       | 0.1       | 513            | 0.0513                | 0.01                 |
| <b>Total</b>       |           |                | <b>45.73</b>          | <b>7.99</b>          |

Table 6.1.4: Electricity cost of the project's hardware resources. [Own creation]

### Internet cost

The basic internet plan costs €21.95 per month [7]; hence, this cost is calculated as follows:

$$\text{Internet cost} = 21.95 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times 122 \text{ days} = \textbf{€89.26}$$

### Office rental cost

The project development always takes place in the FSP office. The FSP office is located in Plaça Francesc Macià, Barcelona, and demands a monthly rental fee of €6,000. Given that there is enough space to accommodate up to 25 employees (which is the number of employees at FSP's Barcelona team, as defined in Section ??), this is the cost calculation:

$$\text{Office rental cost} = 6,000 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times \frac{1 \text{ office}}{25 \text{ employees}} \times 122 \text{ days} = \textbf{€976.00}$$

### Software resources cost

Most software resources are open-source and are available for free. The primary expenses in this cost are Azure and Microsoft resources.

Azure charges a pay-as-you-go fee and offers a customized price for each resource usage [8]. Since the application target user group is small, the most basic/free tier for each

resource is considered. A monthly fee of €3.96 is estimated for this project, as shown in Table 6.1.5.

| <b>Azure resources (€/month)</b> | <b>Cost</b> |
|----------------------------------|-------------|
| SQL DataBase                     | 3.90        |
| Bandwidth                        | 0.01        |
| Blob Storage                     | 0.01        |
| Log Analytics                    | 0.01        |
| Event Grid                       | 0.01        |
| Service Bus                      | 0.01        |
| Functions                        | 0.01        |
| <b>Total</b>                     | <b>3.96</b> |

Table 6.1.5: Azure resource cost. [Own creation]

Accordingly, the total Azure resources cost in the project is derived:

$$\text{Azure resource cost} = 3.96 \text{ €/month} \times \frac{1 \text{ month}}{30 \text{ days}} \times 122 \text{ days} = \mathbf{\text{€24.16}}$$

Based on the Microsoft annual fee of €99 [9], the corresponding total cost is the following:

$$\text{Microsoft resource cost} = 99 \text{ €/year} \times \frac{1 \text{ year}}{365 \text{ days}} \times 122 \text{ days} = \mathbf{\text{€33.09}}$$

Thus, the total software resources cost equals the sum of both costs:

$$\text{€24.16} + \text{€33.09} = \mathbf{\text{€57.25}}$$

### Total generic cost

By summing all previously calculated costs, the total generic cost is obtained, as shown in Table 6.1.6.

| <b>Concept</b>           | <b>Cost (€)</b> |
|--------------------------|-----------------|
| Amortization             | 102.22          |
| Electricity              | 7.99            |
| Internet                 | 89.26           |
| Office Rental            | 976.00          |
| Software                 | 57.25           |
| <b>Generic Cost (GC)</b> | <b>1,232.72</b> |

Table 6.1.6: Total generic cost. [Own creation]

### 6.1.3 Other costs

#### Contingencies

No project strategy can eliminate the possibility of encountering obstacles. Therefore, a time buffer (27 hours) has been included to cover potential setbacks. However, simply adding extra time is insufficient, as it also entails additional resource consumption.

For this reason, a portion of the project budget must be allocated for these potential scenarios. Therefore, the contingency cost amounts to **€1314.57**, which is the 10% of the total general cost calculated so far:

$$PC (\text{€}) + GC (\text{€}) = 11,913.05 + 1,232.72 = \mathbf{13,145.77 \text{ €}}$$

#### Incidental Costs



Incidental costs encompass the project risks analysis outlined in Section ???. The cost of each risk is calculated by extracting its probability, time cost, and required resources from Table ??. A breakdown calculation is presented in Table 6.1.7. Note that the resources required for each risk are identified as Rx (R1, R2, ...).

| Incident                                    | Risk (%) | Time (h) | R1 | R1 Cost (€/h) | R2 | R2 Cost (€/h) | Total Cost (€) | Risk Cost (€) |
|---------------------------------------------|----------|----------|----|---------------|----|---------------|----------------|---------------|
| Coding Bugs                                 | 15       | 10       | D  | 19.87         | HW | 1.99          | 200.66         | 30.10         |
| Tight Project Deadlines                     | 30       | 9        | D  | 19.87         | HW | 1.79          | 180.59         | 54.18         |
| Lack of Experience in Certain Technologies  | 30       | 8        | D  | 19.87         | HW | 1.59          | 160.53         | 48.16         |
| Company's Licenses and Security Permissions | 15       | 0        | D  | 19.87         | HW | 0.00          | 0.00           | 0.00          |
| <b>Total</b>                                |          |          |    |               |    |               |                | <b>132.44</b> |

Table 6.1.7: Incidental cost. [Own creation]

#### 6.1.4 Total cost

After summing the total costs from each section, the final project budget is determined, as shown in Table 6.1.8.

| Concept               | Cost (€)         |
|-----------------------|------------------|
| Personnel Cost (PC)   | 11,913.05        |
| Generic Cost (GC)     | 1,232.72         |
| Contingency Cost (CC) | 1314.57          |
| Incidental Cost (IC)  | 132.44           |
| <b>Total Budget</b>   | <b>14,592.78</b> |

Table 6.1.8: Total cost of the project. [Own creation]

#### 6.1.5 Management Control

Challenges, both expected and unexpected, are always present in any project. Even with a precisely estimated timeline and budget, it is very common for a project to fall short of the initial plan.

To keep the budget and timeline on track, the developer monitors the project through recurring checkpoints. At the end of each week, the actual effort (the real hours) of each completed task is recorded in the project's backlog (Azure DevOps). As mentioned in Section ??, no sprints or milestones are strictly defined; instead, the project follows the KanBan's continuous flow. As a result, it has been decided that at the end of each month, a budget and timeline comparison will be conducted.

To quantify performance, standard Earned Value Management (EVM) will be applied during each monthly review. Cost and schedule efficiency will be measured using the Cost Performance Index (CPI) and the Schedule Performance Index (SPI), respectively. The formulas used to calculate these metrics are defined below.

$$\text{CPI} = \frac{\text{EV}}{\text{AC}} \quad (\text{Earned Value} / \text{Actual Cost})$$

The CPI value can be greater or less than 1, indicating a budget overestimation or underestimation, respectively. If the current expenditure exceeds the budget, some could have requirements listed in Section ?? will be deprioritized and deferred.

$$SPI = \frac{EV}{PV} \quad (\text{Earned Value} / \text{Planned Value})$$

SPI follows the same pattern. Values above 1 reflect that the progress is ahead of schedule, while values below 1 reflect delays. If the second case occurs, key functionalities will be reprioritized with the Product Owner, or the 27-hour buffer will be allocated.

To ensure full traceability, a table comparing estimated and actual costs for each task group is maintained and updated during each monthly review. A sample tracking table for the first monthly monitoring (after week 4 of the Gantt diagram in Section ??) is shown in Table 6.1.9.

| Task Group           | Planned Cost (€) | Actual Cost (€) | Deviation (€)   | CPI         | Status           |
|----------------------|------------------|-----------------|-----------------|-------------|------------------|
| PS                   | 1,433.33         | 1,230.50        | 212.83          | 1.17        | On Budget        |
| PM                   | 2,035.17         | 200.00          | 1835.17         | 10.18       | On Budget        |
| DT                   | 445.87           | 450.00          | -4.13           | 0.99        | Over Budget      |
| FPD                  | 1173.97          | -               | -               | -           | Not Started      |
| Dev                  | 1689.12          | 155.20          | 1,533.92        | 10.88       | On Budget        |
| ER                   | 1,271.47         | 164.30          | 1,107.17        | 7.74        | On Budget        |
| UA                   | 1,271.47         | -               | -               | -           | Not Started      |
| BD                   | 2,384.00         | -               | -               | -           | Not Started      |
| FD                   | 198.67           | -               | -               | -           | Not Started      |
| <b>Total to Date</b> | <b>6,884.95</b>  | <b>2,200.00</b> | <b>4,684.95</b> | <b>6.19</b> | <b>On Budget</b> |

Table 6.1.9: Planned cost, actual cost, deviation, and CPI for each task group. [Own creation]

At the end of the project, the real costs are calculated using the same methodology as the initial budget estimation. With the obtained results, various deviation formulas will be applied in the three areas:

- **Human resources deviation:** To contrast the actual effort with the previous estimated effort of human resources in each task.

$$\text{Human resources deviation} = \sum_{i \in pit} (\text{estimated\_cost\_per\_hour}_i - \text{real\_cost\_per\_hour}_i) \times \text{total\_real\_hours}_i$$

Where  $i$  indexes each individual in  $pit$ , the set of all personnel involved in that task.

- **Amortization deviation:** To compare the previously estimated use of hardware resources with the actual usage time of those resources.

$$\text{Amortization deviation} = \sum_{i \in hr} (\text{estimated\_usage\_hours}_i - \text{real\_usage\_hours}_i) \times \text{price\_per\_hour}_i$$

Where  $i$  indexes each resource in  $hr$ , the set of all hardware resources used in the project.

- **Total cost deviation:** To determine each task's real cost deviation from the estimated cost. The general cost described in the formula represents the sum of the personnel and generic costs, excluding contingency and incident costs. Hence, the following formula is applied to each task:

$$\text{Total cost deviation} = (\text{estimated\_general\_cost} - \text{real\_general\_cost})$$

Note that the resulting total cost deviation value can be either positive or negative. A positive value indicates an overestimation of the initial budget for that resource or task. In this case, the additional budget can be reallocated to address issues that arise during the project. Conversely, a negative value indicates an underestimation, requiring contingency

and incident costs to cover the budget shortfall.

Furthermore, some costs are not included in the deviation calculation. For example, Azure offers a fixed monthly fee; therefore, overusage will not affect the overall cost deviation. The same applies to the remaining omitted costs.

## 6.2 Sustainability

The technological revolution has accelerated dramatically in recent decades. A clear example is the launch of the first telephone (1876), the first computer (1937), the first laptop (1974), the first smartphone (2007), and the first artificial intelligence (2017) [10].

This acceleration pace reveals that the time it takes for a technological invention to emerge and significantly impact society continues to shrink. Consequently, this exponential growth underscores the increasing importance of evaluating the economic, environmental, and social impacts of technological projects. These three dimensions together constitute sustainability, the topic discussed in this section.

Building on this trend, technology will continue shaping our future in the coming years. The primary mission of engineers is to make people's daily lives easier. However, developing a solution is not enough; a study of its sustainability effects is also a must.

Therefore, this section addresses sustainability questions, enabling a better evaluation of the project's impact across the economy, environment, and society. This analysis raises awareness of the weight of sustainability in project development.

Personally, the author recognizes limited sustainability knowledge, mainly covering environmental footprint and social impact. The economic dimension and multiple sustainability indicators require further study. The subsequent subsection answers a series of key sustainability questions within the author's current expertise.

### 6.2.1 Economic Impact

#### **Regarding PPP: Have you estimated the project execution cost?**

The project budget is €14,592.78, as detailed in Section 6.1. This corresponds to the full "bill" for the application, including all personnel, hardware amortization, software licensing, and contingency costs.

#### **Regarding Useful Life: How are currently solved economic issues (costs...) related to the problem that you want to address (state of the art)?**

According to the problems stated in Section ??, the current time-consuming, low-efficiency expense management process delivers minimal value while misallocating company personnel costs for both employees and finance staff.

#### **Regarding Useful Life: How will your solution improve economic issues (costs, etc.) with respect to other existing solutions?**

The primary benefit of the proposed solution is the optimization of operational efficiency. The current manual process requires approximately 225 hours per year, as estimated in Section ?. The application is designed to reduce the time required for this activity by at least 75%, in accordance with the non-functional requirement NFR6 from Section ?.

Based on this reduction, the solution is expected to generate annual cost savings of approximately €5,062.5 (derived from Section ??). Considering the estimated development cost, the Return On Investment (ROI) is projected to be achieved within 3 years. This demonstrates that the proposed solution is more economically viable than investing in external solutions, supporting its adoption as a sustainable long-term approach.

However, an initial budget misestimation could extend the ROI period. This is mitigated with the 27-hour built-in buffer (calculated in Section ??) and the MoSCoW prioritization strategy (employed in Section ??). As a result, a high-value product is most likely to be delivered, since could have strategies can be postponed in case of budget shortfalls.

## **6.2.2 Environmental Impact**

### **Regarding PPP: Have you estimated the project's environmental impact?**

The only resource directly affecting the environmental footprint is energy consumption. As shown in Table 6.1.4, the total energy consumption is 45.73 kWh. Using a conversion factor of 0.283 kg CO<sub>2</sub>/kWh [11], the project's carbon footprint is around 12.94 kg CO<sub>2</sub>.

### **Regarding PPP: Have you considered minimizing its impact, for example, by reusing resources?**

The project's environmental impact is minimized by conducting the full project development in FSP's shared office, reusing existing lighting and preventing unnecessary energy consumption.

### **Regarding Useful Life: How is the problem that you want to address currently solved (state of the art)?**

The current employees' annual effort spent on manual expense management is 225 hours, corresponding to an estimated energy consumption of 12.63 kWh (assuming a PC power consumption of 65 W). As the proposed solution reduces the required effort by 75%, the associated energy consumption is expected to decrease by around 3.16 kWh. In addition, digitalizing the process reduces the environmental impact associated with printing, transportation, and the physical handling of expense records.

In addition, the application uses Azure for software resources. The Azure resources required for application development cost less than traditional physical resources. This cost optimization results from significantly reduced resource consumption, as cloud resources eliminate the need for transportation and frequent physical replacements. Updates and fixes are performed virtually, further reducing hardware requirements and electronic waste.

Nevertheless, suboptimal code may lead to excessive CPU and memory consumption, resulting in a negative environmental impact on the project. Therefore, adherence to high-quality coding practices, as defined in subobjective SO5 in Section ??, is emphasized throughout the project development. As a result, a more sustainable application is delivered, using only essential resources and minimizing overall consumption throughout its lifecycle.

### **Regarding Useful Life: How will your solution improve the environment with respect to other existing solutions?**

This application is tailored for the company's exclusive use and excludes any irrelevant

functionality from the final product. It streamlines resource consumption by avoiding the unnecessary computation performed by other existing applications.

### **6.2.3 Social Impact**

#### **Regarding PPP: What do you think you will achieve—in terms of personal growth—from doing this project?**

This project enables the author to apply prior academic knowledge and software development skills in a real-world context. As a result, it enriches their working experience as future software engineer. Furthermore, it is the author's first experience managing and leading an entire project independently, strengthening self-learning, self-management skills and the ability to build software projects from scratch.

#### **Regarding Useful Life: How is the problem that you want to address currently solved (state of the art)?**

The current expense management process is handled manually through traditional methods. This involves manually entering data into Excel spreadsheets, emailing them to the finance department, and reviewing them line by line.

#### **Regarding Useful Life: How will your solution improve the quality of life (social dimension) with respect to other existing solutions?**

The proposed solution optimizes time spent on expense management while reducing employees' frustration and fatigue. This frees employees from this tedious task, allowing additional time to focus on more meaningful work. As a result, it maximizes employees' productivity.

However, negative effects are also encountered with the proposed application. An initial resistance to change and a steep learning curve are expected. In addition, a non-intuitive UI would exacerbate the adoption process. To mitigate the last risk, the Product Owner is actively involved in task DT5, a user manual guide is required as part of task PM4 (both defined in Section ??), and compliance with NFR2 from Section ?? is ensured.

Another potential risk is excessive reliance on AI extraction, leading employees to poorly review the AI-extracted data, increasing the risk of errors. To mitigate this, the workflow is redesigned so that reports cannot be submitted until employees have explicitly reviewed the extracted information.

#### **Regarding Useful Life: Is there a real need for the project?**

Regarding all project sustainability impacts (social, environmental, and economic), there is a clear need to build a robust solution. This solution overcomes challenges from all involved parties while introducing a sustainability-focused approach.

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